

EXAM 7 – SPRING 2012

19. (2.5 points)

ABC Company is a small personal lines insurance company. ABC is lightly capitalized relative to its written premium. The company's CFO worries that the rating agencies might downgrade the company if it experiences extremely adverse underwriting results in the upcoming accident year. ABC Company has received quotes for stop-loss reinsurance programs from two reinsurance companies. The CFO is considering the following options.

<u>Option</u>	<u>Reinsurer's rating</u>	<u>ABC's expected return on GAAP equity</u>
Reinsurer X	AAA	10%
Reinsurer Y	A– with negative outlook	15%
No reinsurance		25%

a. (1.5 points)

Discuss the CFO's reinsurance options with regard to the costs of financial distress and taxation.

b. (1 point)

Discuss, in terms of agency theory, a CFO bonus structure for ABC Company in which the CFO's bonus is maximized by achieving a 30% return on GAAP equity and minimized by achieving less than a 15% return on GAAP equity.

CONTINUED ON NEXT PAGE

Question 19 Sample Answer

Solution 1

- a) If the insurance company is downgraded it could incur significant costs trying to raise new capital, which is already thin. High yield bond issues or issuance of discounted stock are not preferred. It should go with reinsurer x to prevent volatility in its U/W results. While its expected ROE will be less, it will be more stable. This in turn will reduce potential tax effects, as excessive income in one period will be taxed at a higher bracket than if it were stable overall years. With no reinsurance or the reinsurance that produces a downgrade, the ROE may be higher but the volatility could result in risk of insolvency or financial distress costs.
- b) If the CFO's bonus is maximized with a 30% ROE, the company is basically rewarding the CFO to take risks with the shareholders money. Under agency theory you want to align the interests of principals (shareholders, and the agents, the CFO) who you put in the charge of your money for your natural gain. The CFO will not want to mitigate the risk because it will lower his bonus as the ROE declines. This may result in financial distress costs for the shareholders if the insurer is downgraded and needs to raise capital.

Solution 2

- a) Financial distress is expensive: if debts (bonds) are issued, they will be at a high interest rate. If stock is issued, it will be at a discount. May need to quickly sell assets, but the sale will be at depressed prices. Having reinsurance reduces the chance of being placed in this situation.

Having stable & predictable results will minimize taxes.

– Reins x has a great rating (low credit risk) X seems to be more stable, therefore choose this reinsurance.	y has moderate rating	w/ no reins, a large loss may cause distress if not enough capital available.
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- b) CFO acts on behalf of the principal for mutual benefit. A compensation setup like this will incentivize the CFO to be riskier and purchase no reinsurance (since that option has a 25% expected return). Stakeholders (especially policyholders) would not like this. They prefer stable, predictable outcomes. P/h's are willing to pay more from a more stable insurer, & demand a discount on premium if it's not stable. The uncertainty of future indemnification may also scare customers away.

If the CFO were aware of this, he may choose some reinsurance as opposed to more and receive a minimal bonus instead of being out of a job.

Examiner Comment

For the a. part, candidates would have to discuss at least 3 costs of financial distress. Credit would also be given to mentioning the following:

- Reduction in earnings and market share
- Policyholders are less willing to pay as much for a financially distressed insurer

Common errors included the following:

- Just saying that the firm would experience a greater chance financial distress without reinsurance, or saying that financial distress is costly, without mentioning what these costs are.
- Saying that a firm would be downgraded when in financial distress. This was already mentioned in the question as a concern for the company's CFO.
- Saying that a firm would pay lower taxes because of the costs of reinsurance
- If a candidate said that a firm would pay lower taxes with reinsurance because of a lower expected ROE, .25 points were deducted. While it is technically correct, it does not fully explain why the expected ROE is lower (because of greater earnings stability).

For the b. part candidates also got credit for saying that the CFO is the agent to the policyholders and bondholders (though in this last case it wouldn't be a direct relationship). Candidates also got credit for saying that the interests of the CFO and the principals are misaligned. Common errors included the following:

- Not mentioning the relationship between the CFO (agent) and the shareholders/policyholders/debtholders (principals). This is a key concept of agency theory.
- Not discussing the conflict of interest between the CFO and the principals. This is another key concept of agency theory.
- Saying that the CFO was the agent to the company. The company is not a principal.
- Not specifically discussing the interest of the principals. For example, saying that the CFO was putting the company at risk without specifically saying that the objective of the principal was to minimize risk.